CHEN GUANHUA

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HIGHLIGHTS

- **Publications:** 7 first-author papers at top-tier NLP venues, including 3 ACL (main & findings), 1 IEEE/ACM TASLP, 1 KBS, and 1 LREC-COLING; 4 manuscripts under review, 2 in preparation; 1 patent pending.
- **Research:** Reinforcement Learning for Personalized LLMs, agentic systems for scientific discovery, and cross-modal alignment between LLMs and EEG signals for affective/cognitive modeling.
- **Projects and Systems:** Built 5 AI systems for real-world applications, including an AI research assistant, a personalized education platform, and a multi-agent system for personal travel planning.
- **Mentorship:** Mentored 7 MSc and BSc students as primary supervisor; 2 advisees received Best Final Year Project Awards, with co-authored publications arising from the collaborations.
- **Awards:** 7 competition awards, including Winner Prize of 2021 Global Open Data Application Competition (Legal QA Track), and 3rd Prize, CCIR-Cup National Information Retrieval Challenge Cup.

EDUCATION

PhD in Computer Science, University of Macau — Expected 2027
PhD Candidate in NLP^{2}CT Lab. Supervisor: Derek F.Wong
Research Area: Multi-modal RAG, Personalized LLMs, LLM Agents

Master in Computer Science, University of Macau — 2021-2023
Research Area: Question Answering, Dialogue System

Bachelor in Computer Science, University of Macau — 2016 - 2020
Minor Major: Mathematical application

PUBLICATIONS

Chen, G., Zhan, R., Wong, D. F., & Chao, L. S. (2023). Multi-level curriculum learning for multi-turn dialogue generation. IEEE/ACM Transactions on Audio, Speech, and Language Processing, 31, 3958-3967.

Chen, G., Zhan, R., Wong, D. F., & Chao, L. S. (2024). Dynamic curriculum learning for conversation response selection. Knowledge-Based Systems, 293, 111687.

Chen, G., Yao, Y., Wong, D. F., & Chao, L. S. (2024, May). A two-stage prediction-aware contrastive learning framework for multi-intent NLU. In Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024) (pp. 1778-1788).

Chen, G., Yao, Y., Gao, C. J., Chao, L. S., Wan, F., & Wong, D. F. (2025). Not all lora parameters are essential: Insights on inference necessity. arXiv preprint arXiv:2503.23360.

Chen, G., Yao, Y., Chao, L. S., Liu, X., & Wong, D. F. (2025, July). SGIC: A Self-Guided Iterative Calibration Framework for RAG. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (pp. 28357-28370).

Yan, W., Wu, J., **Chen, G.**, Su, S., Shen, P., Wong, D. F., & Zhou, W. H. (2025). Standardization and interpretable analysis of geological database using retrieval-augmented large language model. Geodata and AI, 100047.

ACADEMIC PAPERS

Guanhua Chen*, Chuyue Huang*, Yutong Yao, et al. From Scenes to Elements: Multi-Granularity Evidence Retrieval for Verifiable Multimodal RAG. *ACL 2026* (Accepted to Findings)

Guanhua Chen*, Yutong Yao*, Shenghe Sun, et al. Chain-of-Procedure: Hierarchical Visual-Language Reasoning for Procedural QA. *ACL 2026* (Accepted to Findings)

Guanhua Chen, Yang Xu, Yutong Yao, et al. AMUSE: An Agentic Platform for Managing the Lifecycle of Evolving Research Ideas. *ACL 2026 Demo.* (Under Review)

Jiameng Lu, Chuyue Huang, **Guanhua Chen**, et al. AdaptiMath: A Controllable Framework for Interpretable Multimodal Math Tutor Evaluation. *TACL.* (Under Review)

Shaoze Su, Wei Yan, Peizhong Yu, **Guanhua Chen**, Derek F.Wong, Hannah Zhou. Light Knowledge-Informed LLM Agent for Construction Site Characterization. *Advanced Engineering Informatics.* (Under Review)

Guanhua Chen*, Liulei Zhang*, Yutong Yao, et al. ADMIRE: A Dual-Chain Multi-Agent Framework for Idea Novelty Assessment. *ARR.* (Under Review)

Yutong Yao, **Guanhua Chen**, et al. Perceiving More, Reasoning Less? Benchmarking Fine-grained Cross-Image Transformation in VLMs. (In Preparation)

Guanhua Chen, CI-JUN GAO, Yutong Yao, et al. Can Large Language Models as Proxies for Human Cognition in Cross-Modal Emotional Conflicts? (In preparation)

PROJECTS

iSuperviz. An AI research assistant that tracks new papers, organizes references, identifies research overlaps to prevent scooping, and detects AI hallucinations in academic writing. Designed to streamline the research workflow for academics and graduate students. (Try it here)

RapiLearn AI. A personalized learning platform that generates course content, exercises, and mind maps from any requirements or PDF uploads. Integrated an AI tutor system that analyzes individual student weaknesses and provides teachers with data-driven recommendations for personalized instruction. (Try it here)

UUTrip. Built a multi-agent system that generates detailed travel itineraries based on a large knowledge graph of Macau attractions. Agents collaborate to create personalized travel plans tailored to user preferences. (Try it here)

Macau LLM. Built a localized QA system powered by a semi-automated Macau database and multi-granularity hierarchical retrieval pipeline. Enables accurate responses to Macau-specific queries from large-scale data. Currently under active development and optimization.

Portuguese and Chinese Translation System. Collected and cleaned over 3 million Chinese-Portuguese parallel corpus, with special optimization for Macau-localized content. Trained a Transformer-based Chinese-Portuguese translation model that was deployed for practical use by relevant government departments.

PATENT

- **Guanhua Chen**, Hui Huang, et al. "A Trip Planning Method and Apparatus", 2025114588106, Pending.

COMPETITION AWARDS

- The First Prize of APP Design Competition of University of Macau in 2019
- Third Prize of MathorCup Mathematical Modeling
- IEEE Macau Industrial Electronics Society (IES) Undergraduate Contest: Second Prize, Macau
- IEEE Project Competition 2020 (Macau): Third Prize, Macau
- Pan-Pearl River Delta Region Universities IT Project Competition: Third Prize, Macau
- 2021 CCIR-Cup National Information Retrieval Challenge Cup, Third Prize
- 2021 Global Open Data Application Innovation Competition, Legal QA Track, Winner Prize

SUPERVISION EXPERIENCE

- **Yutong Yao (MSc):** Built a multimodal reasoning dataset to evaluate LLMs' multi-granularity reasoning capabilities across images.

- **Chuyue Huang, Jiameng Lu (MSc):** Developed a math education benchmark dataset and evaluation framework to assess existing AI Tutor systems.
- **Jingdong Yang (MSc):** Designed a multi-agent collaborative AI Tutor framework that delivers personalized education through concept explanation, adaptive testing, and guided Socratic questioning.
- **Tianchang Song (MSc):** Predicted future citations via in-context learning, leveraging multi-dimensional similarity between emerging ideas and existing literature.
- **Huang Chuyue, Song Xueqing (FYP):** Macava: An MLLM-Based Macau Tourism Assistant. **Best Final Year Projects, FST-UM (2025)**
- **Jiang Rui, Nan Shuyuan (FYP):** WeSpeak: Web Based Simultaneous Interpretation System. **Best Final Year Projects, FST-UM (2023)**
- **Yao Yutong (FYP):** UMor-PG - The Intelligent Chatbot for Postgraduate Admission Information!. **The Dean's Final Year Project List, FST-UM (2022)**

RESEARCH EXPERIENCES

Macao Tourism Planning Agent **2024-2025**
Type: FDCT Project; Role: Leader
Built a knowledge graph based on local Macao data and designed an agent framework to generate personalized travel itineraries aligned with user preferences (time, style, etc.).

Question Generation from Auto Repair Manuals **2021-2022**
Type: FDCT Industry-Academia Collaboration Project; Role: Researcher
Performed OCR and structured formatting on Lexus user manuals, fine-tuned a question generation model based on BART for QA data augmentation, improving response efficiency of the QA system.